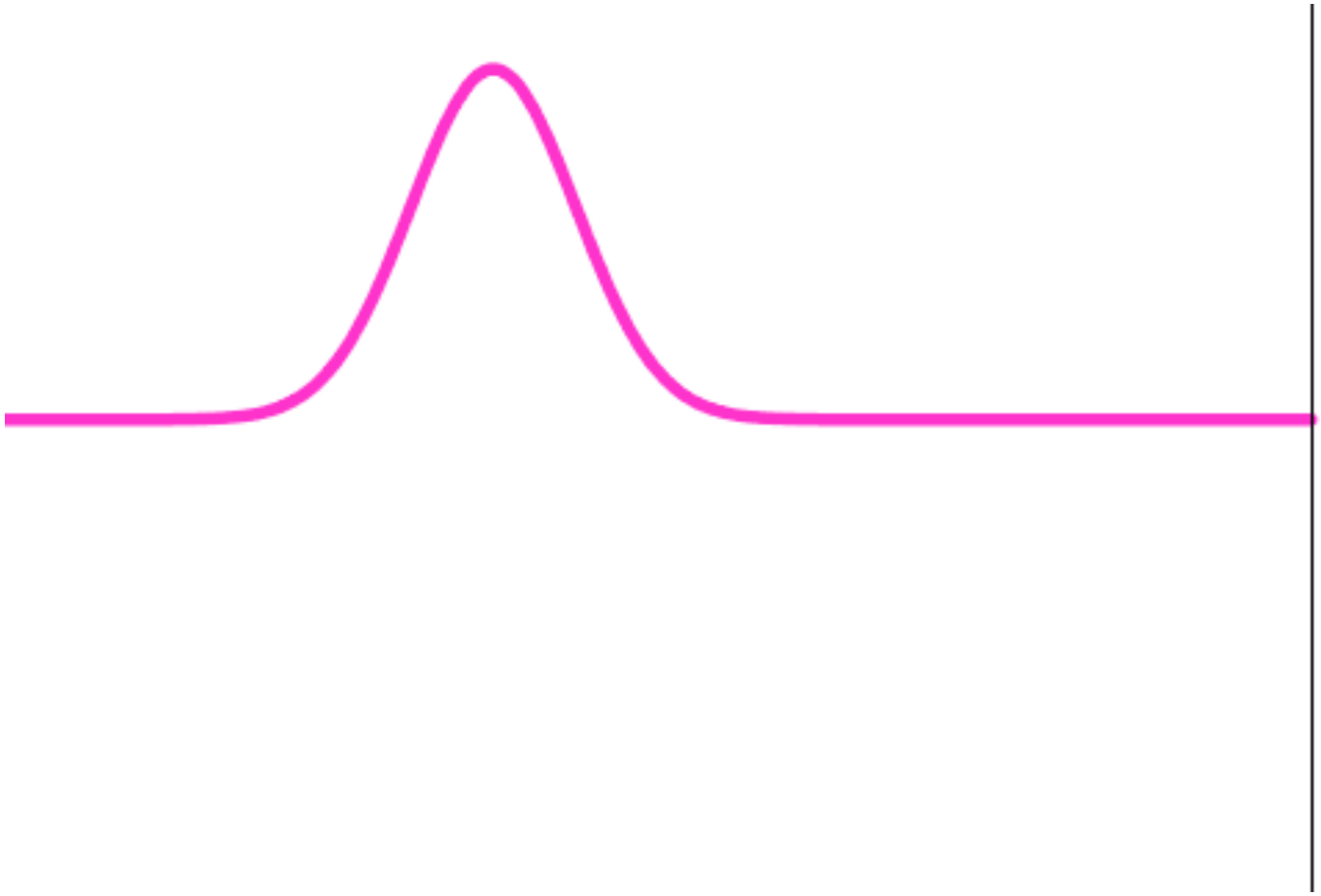
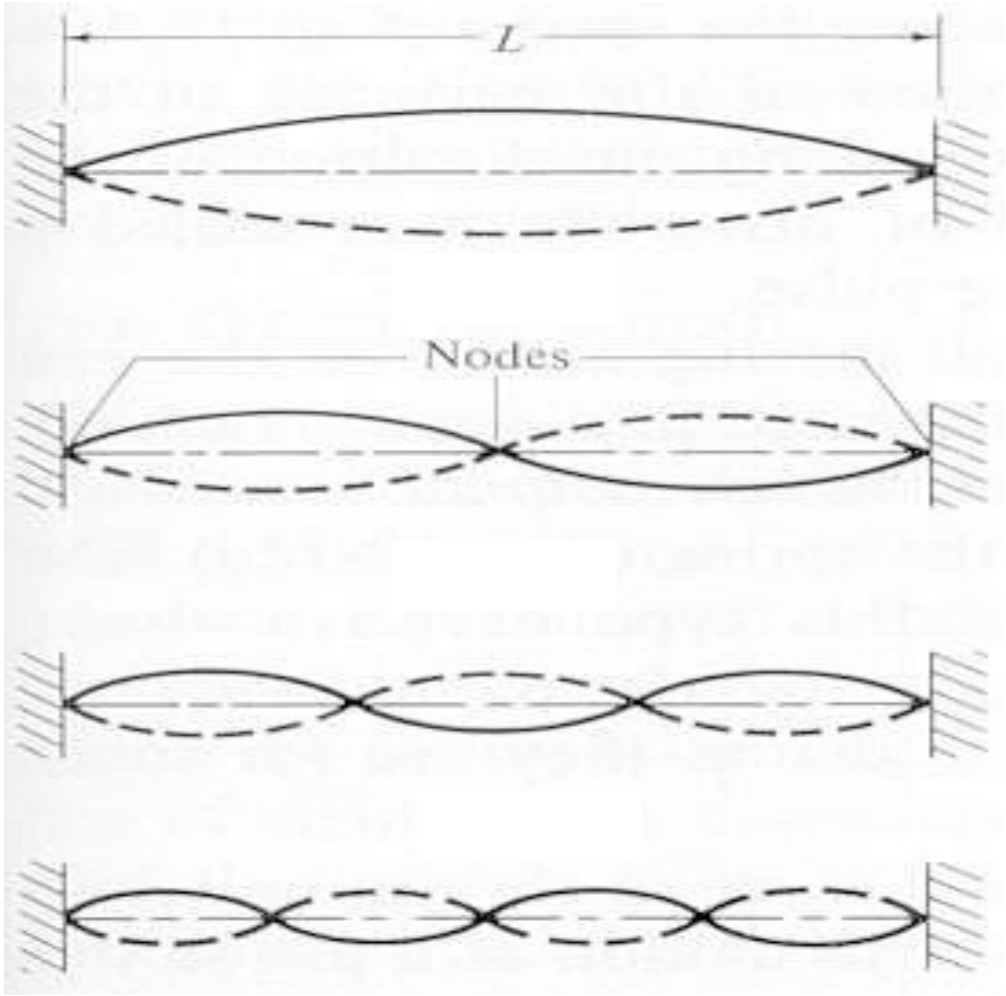
Reflection at a rigid boundary: π phase shift



https://www.ophysics.com/waves/waves3.html

Resonant modes of a string



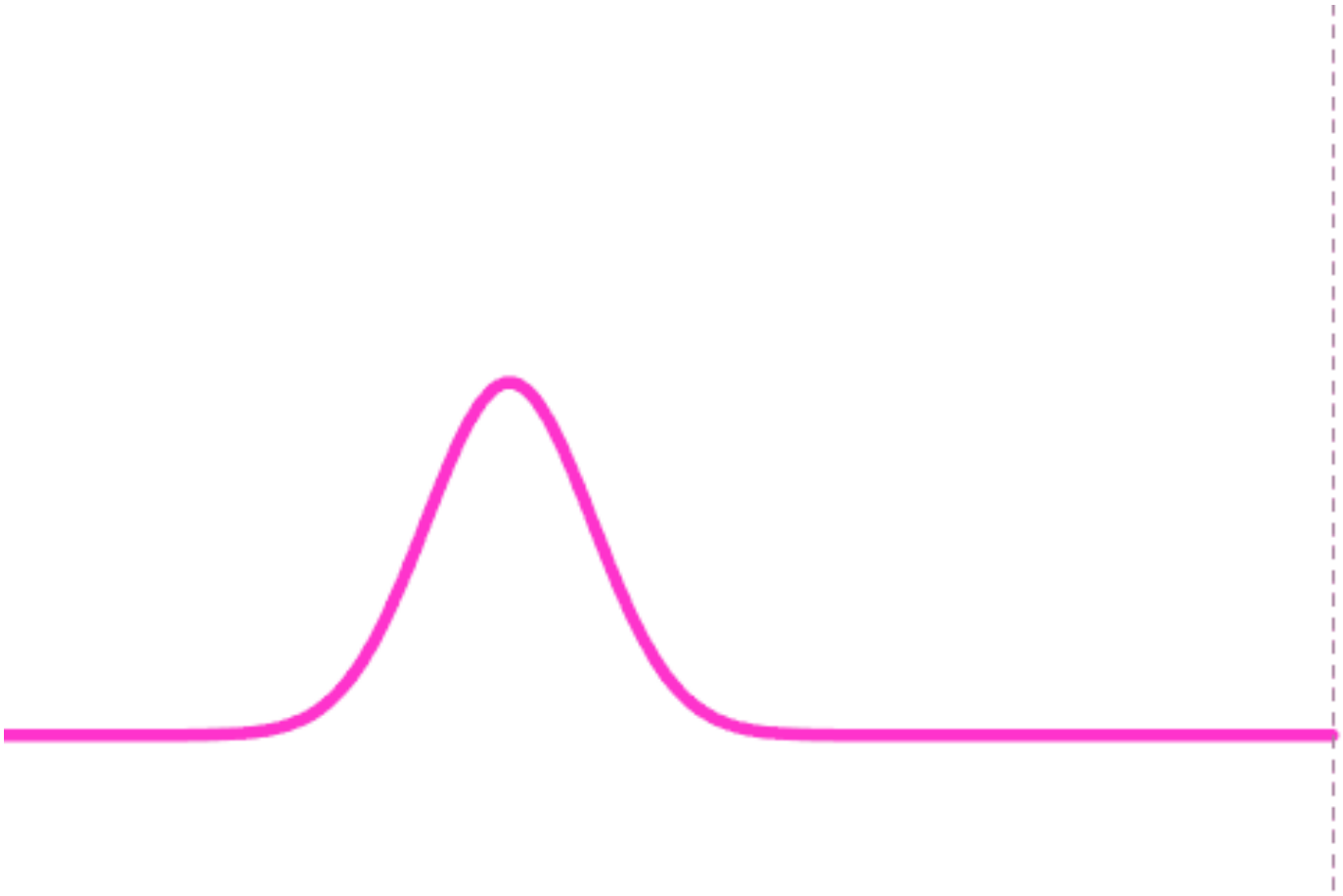
$n = 1$ has 2 nodes and 1 antinode

$n = 2$ has 3 nodes and 2 antinodes

$n = 3$ has 4 nodes and 3 antinodes

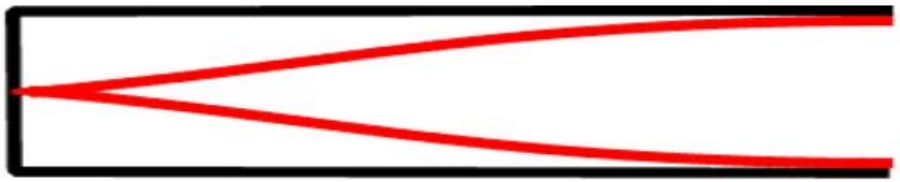
$n = 4$ has 5 nodes and 4 antinodes

Reflection at an open end: no phase shift

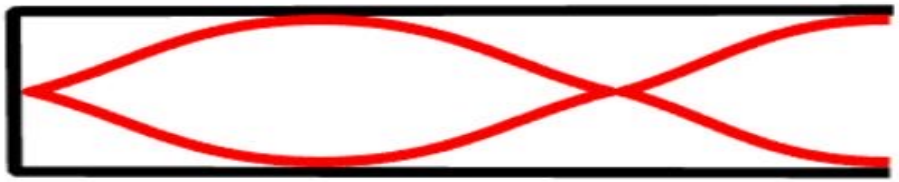


https://www.ophysics.com/waves/waves3.html

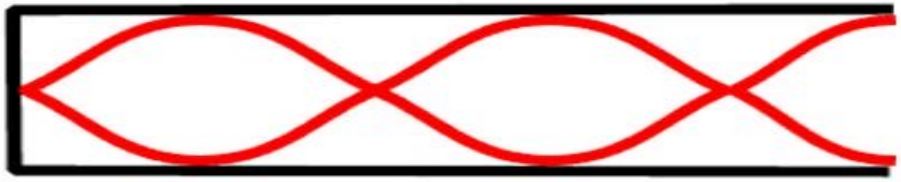
Air column: one end open, one closed



1st harmonic has 1 node at fixed end and 1 antinode at open end



2nd harmonic has 2 nodes and 2 antinodes

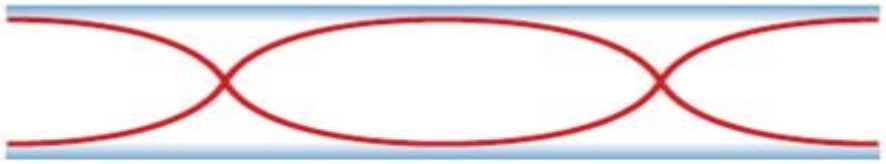


3rd harmonic has 3 nodes and 3 antinodes

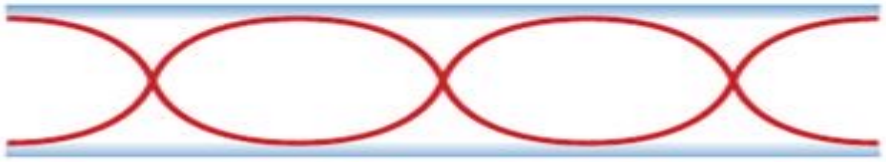
Air column: both ends open



1st harmonic has 1 node in middle and 2 antinodes at open ends



2nd harmonic has 2 nodes and 3 antinodes



3rd harmonic has 3 nodes and 4 antinodes



https://www.youtube.com/watch?v=52UAEQfMTtU